

Riley Harper

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EDUCATION

University of North Carolina at Chapel Hill December 2025
B.S. Statistics and Analytics, B.S. Applied Mathematics, Minor in Entrepreneurship
Coursework: Deep Learning, Machine Learning, Reinforcement Learning, Stochastic Modeling, Optimization, Linear Algebra, Discrete Math, Differential Equations, Data Structures and Algorithms, Nonlinear Dynamics, Fluid Mechanics, Real Analysis

SKILLS

- **Languages:** Python, Java, SQL, C/C++, R, MATLAB, Julia, Wolfram Language, C#, JavaScript, HTML/CSS
- **Technologies:** AWS (Bedrock, Lambda, S3), Azure; Docker, Git, Kubernetes, REST APIs; GitHub Actions, Jenkins; PyTorch, TensorFlow/Keras, scikit-learn, HuggingFace, DeepLabCut; NumPy, Pandas, dplyr, yfinance; BeautifulSoup, Selenium; React, Unity

EXPERIENCE

National Institutes of Health (NIH) January 2025 – Present
Research Intern

- Developed CNN/GNN for multi-animal pose estimation in Neurobiology lab with DeepLabCut, SLEAP, PyTorch on HPC cluster.
- Packaged codebase for Python Package Index (PyPI) release and added unit tests, reducing user setup time from hours to minutes.

AI@UNC August 2023 – Present
Technical Chair & Co-Founder

- Co-founded Carolina’s inaugural faculty-mentored AI research club, growing membership to 400+ students and mentoring students involved in 20+ faculty-led research projects in NLP, CV, GANs. Led workshops on PyTorch, GitHub, AWS, BIDS, HPC clusters.
- Head Developer for NeuroSpark, integrating AI & VR for awake craniotomy with Unity and PyTorch; secured \$25k+ grant funding.

Velocity Labs August 2024 – Present
Quantitative Research Analyst

- Discussed HFT and options strategies for student-managed fund with \$200k+ AUM, outperforming 98.89% of retail investors.
- Selected at a 3.3% acceptance rate from 500+ applicants and developed a Kalshi arbitrage trading algorithm with 600%+ returns.

Fidelity Investments June 2025 – August 2025
Machine Learning Engineering Intern

- Containerized and deployed SageMaker workloads with CI/CD pipelines, using Jenkins, Docker, and Kubernetes.
- Wrote first code in production for the AI Portal, a platform unifying data scientists (DS) and software engineers (SWE) best practices.
- Engineered integration tests utilizing MLOps/DevOps principles for Model Maker platform, catching two bugs before release.
- Worked on GenAI Gateway, an abstraction layer uniting AWS Bedrock and Azure for secure, enterprise-grade LLM access at scale.

Artificial Intelligence Center of Czech Technical University (ČVUT) July 2024 – August 2024
Research Intern

- Researched Bayesian network optimization at Czechia’s top computer science school while studying abroad in Prague.
- Competed in department Quantum Computing Hackathon, building 2D quantum walk algorithm capable of solving mazes in $O(\sqrt{n})$.

Chapel Thrill Escapes January 2022 – April 2024
Chief Executive Officer (CEO)

- Raised \$40k+ and led 70+ students to launch the nation’s first student-run escape room, later expanding to a second flagship Franklin Street venue, rebuilt the website, and coded an in-house MailChimp/Salesforce-style CRM.
- Produced pro-forma financials, annual reports, and forged partnerships with Carolina Housing, 1789 Fund, BeAM Makerspace.

Eats2Seats September 2021 – October 2023
Site Lead

- Scaled startup to \$2M in annual revenue, expanding to three new markets (Chicago, Madison, Baton Rouge), training 10k+ event staff, and setting a Dean Dome single-game sales record of \$180k+.

RESEARCH

Co-Editor-in-Chief for the UNC Journal of Undergraduate Research December 2023 – Present
Machine Learning Research Fellow for the UNC School of Medicine August 2024 – September 2025
Computer Vision Directed Reading Program January 2025 – April 2025
Machine Learning Undergraduate Research Assistant for BRANE Lab August 2023 – September 2024
Cryptocurrency Undergraduate Research Assistant for Rethinc. Labs May 2023 – August 2023

AWARDS

Secured \$100k+ in grants, prizes, and scholarships

- Jane Street Estimathon (1st x2), CDC Datathon (1st), Duke DataFest (Finalist), VC Competition (1st), 1789 Student Venture Fund (x3), Map the System (Global, National, and UNC Finalist x2), NIH Academic Internship Program IRTA, NSF Research Training Grant (x2), NeuroSpark Research Grant, UNC Parent’s Council Grant, Food Insecurity Grant, Oxford Highly Commended Research Award, Dennis Gillings Innovation Fund, Trentini Scholarship, Gold Presidential Service Award, UNC Leader of the Month Award